

	A	B	C	D	E	F	G	H	I
2	total sell basic	560	<=	600		basic	xp	vxp	
3	total sell xp	1200	<=	1200		# produce	560	1200	0
4	total sell vxp	0	<=	50		price	316	450	560
5						cost	150	225	275
6	assembly	10000	<=	10000		assembly	5	6	8
7	testing	2960	<=	3000		testing	1	2	3
8									
9	total sell	716960				labor price	11		
10	component cost	354000				labor test	15		
11	assembly cost	110000							
12	labor cost	44400							
13	total profit	208560							

Decision variables

x_1 = # of Basic models to be produced next month
 x_2 = # of XP models to be produced next month

Objective function Maximize $z = 80x_1 + 129x_2$

Constraints: $5x_1 + 6x_2 \leq 10000$ $x_1 + 2x_2 \leq 3000$
 $x_1 \leq 600$ $x_2 \leq 1200$ $x_1, x_2 \geq 0$

Set Objective: \$I\$15

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

\$G\$3:\$I\$3

Subject to the Constraints:

\$B\$2:\$B\$4 <= \$D\$2:\$D\$4

\$B\$6:\$B\$7 <= \$D\$6:\$D\$7

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$G\$3	# produce basic	560	0	96	11.5	1
\$H\$3	# produce xp	1200	0	129	1E+30	13.8
\$I\$3	# produce vxp	0	-1.6	152	1.6	1E+30

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$B\$2	total sell basic	560	0	600	1E+30	40
\$B\$3	total sell xp	1200	13.8	1200	50	33.33333333
\$B\$4	total sell vxp	0	0	50	1E+30	50
\$B\$6	assembly	10000	19.2	10000	200	2800
\$B\$7	testing	2960	0	3000	1E+30	40

Sensitivity analysis: Allowable Increase & decrease

How much coefficient could change before optimal solution would change.

- If Basic unit margin is between (80-80, 80+27.5) = (0, 107.5), the optimal # to produce Basic remains the same
- If XP unit margin is between (129-33, 129+∞) = (96, ∞), the optimal # to produce XP remains the same.

A	B	C	D	E	F	G	H
6	Variable Cells						
7							
8	Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
9	\$B\$16	Number to produce Basic	560	0	80	27.5	80
10	\$C\$16	Number to produce XP	1200	33	129	1E+30	33

Sensitivity Analysis: Reduced cost

- The reduced cost for decision variable with its upper bound indicates how much worse its coefficient must be before it will decrease from its upper bound (UB).
- The optimal number to produce XP is at its UB (1200 units). If XP margin decreases more than 33, then it is no longer optimal to produce at UB.
- The reduced cost for decision variable with value 0 indicates how much better that coefficient must be before that variable enters at a positive level.
- The reduced cost for any variable between 0 and its upper bound in the optimal solution is irrelevant.

Sensitivity Analysis: Shadow Price

- The shadow price indicates the change in the optimal value of the objective when the right side of some constraint changes by one unit.
- Decreasing marginal effect
- An inequality constraint is **binding**, if the solution makes it an equality.
- Otherwise, it is called **non-binding**.
- The shadow price on a non-binding constraint is zero.

Complementary slackness condition

$$\lambda_1(10000 - (5x_1 + 6x_2)) = 0$$

$$\lambda_2(3000 - (x_1 + 2x_2)) = 0$$

Infeasibility and Unboundedness

A solution is *feasible* if it satisfies all of the constraints.

It is possible that there are no feasible solutions to the model. It could happen when:

- There is a mistake in the model (an input was entered incorrectly).
- The problem has been so constrained that there are no solutions left.

In general, there is no foolproof way to remedy the problem of infeasibility.

Another problem occurs if the objective is unbounded – that is, it can be made as large or small as you like.

- When this happens, Solver states that the objective cell does not converge.

linear, piecewise linear, or nonlinear?

Identify the decision variable(s). Then explain your answer in one sentence.

- Average ordering cost
 - An order costs $K = 100$ THB.
 - If $x > 0$ units are ordered, then the average ordering cost per unit is $A(x) = \frac{100}{x}$.
- Revenue with fixed prices
 - Prices p_1, p_2 are known constants, while x_1, x_2 are production quantities.
 - Total revenue is $R(x_1, x_2) = p_1x_1 + p_2x_2$.
- Revenue when price is a decision
 - Demand is $d(p) = 100 - 10p$, so revenue is $R(p) = p d(p) = p(100 - 10p)$.
- Shipping with quantity tiers
 - The first 100 units cost 5 THB per unit; additional units cost 2 THB per unit.
 - Cost $C(q) = 5 \min(q, 100) + 2(q - 100)^+$.
 - Is $C(q)$ linear in q ?
 - [Optional] Can it be represented linearly using two shipment variables?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
2		model 1 model 2 model 3 model 4 model 5 model 6 model 7 model 8												
3	line_1	1500	0	0	125	0	0	1000	0	0	total_assembly	20000	<=	20000
4	line_2	0	0	0	475	0	1000	0	0	0	total_labor_line_1	5000	<=	5000
5	total	1500	0	0	600	0	1000	1000	0	0	total_labor_line_2	4187.5	<=	4000
6		<=	<=	<=	<=	<=	<=	<=	<=	<=	cost_assembly	11		
7		1500	1250	1250	1250	1000	1000	1000	800	0	cost_line_1	19		
8	assembly	4	5	5	5	5.5	5.5	5.5	6	6	cost_line_2	17		
9	labor_line_1	1.5	2	2	2	2.5	2.5	2.5	3	3	total_profit	61582.5		
10	labor_line_2	2	2.5	2.5	2.5	3	3	3.5	3.5	3.5				
11	component	150	225	225	225	250	250	250	300	300				
12	sell	350	450	460	470	500	525	530	600	600				
13	total_cost_assembly	44	55	55	55	60.5	60.5	60.5	66	66				
14	total_cost_line_1	28.5	38	38	38	47.5	47.5	47.5	57	57				
15	total_cost_line_2	34	42.5	42.5	42.5	51	51	51	59.5	59.5				
16	margin_line_1	127.5	132	142	152	142	167	172	177	177				
17	margin_line_2	122	127.5	137.5	147.5	138.5	163.5	160	174.5	174.5				
18	profit	191250	0	0	89062.5	0	163500	172000	0	0				

- Let $n = 2$ be the number of decision variables
- Let $m = 4$ be the number of constraints
- Let a_{ij} be the coefficient of decision variable x_j in constraint i (row i and column j)

The LP can be written as

$$\max \sum_{j=1}^n c_j x_j$$

$$\text{s.t. } \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m,$$

$$x_j \geq 0, \quad j = 1, \dots, n.$$

Objective function becomes $\sum_{j=1}^n c_j x_j = c_1 x_1 + c_2 x_2 = 80x_1 + 129x_2$

Constraint $i = 1: \sum_{j=1}^2 a_{1j} x_j = a_{11} x_1 + a_{12} x_2 = 5x_1 + 6x_2 \leq 10000$

Sensitivity analysis: Reduced cost (review)

- The reduced cost for decision variable with its upper bound indicates how much worse its coefficient must be before it will decrease from its upper bound (UB).
- The reduced cost for decision variable with value 0 indicates how much better that coefficient must be before that variable enters at a positive level.
- The reduced cost for any variable between 0 and its upper bound in the optimal solution is irrelevant.

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$19	Number tested on line 1 Model 1	1500	0	127.5	1E+30	2.125
\$C\$19	Number tested on line 1 Model 2	0	-20	132	20	1E+30
\$D\$19	Number tested on line 1 Model 3	0	-10	142	10	1E+30
\$E\$19	Number tested on line 1 Model 4	125	0	152	2.833333333	1.7
\$F\$19	Number tested on line 1 Model 5	0	-25.875	142	25.875	1E+30
\$G\$19	Number tested on line 1 Model 6	0	-2.125	167	2.125	1E+30
\$H\$19	Number tested on line 1 Model 7	1000	0	172	1E+30	4.125
\$I\$19	Number tested on line 1 Model 8	0	-6.75	177	6.75	1E+30
\$B\$20	Number tested on line 2 Model 1	0	-2.125	122	2.125	1E+30
\$C\$20	Number tested on line 2 Model 2	0	-20	127.5	20	1E+30
\$D\$20	Number tested on line 2 Model 3	0	-10	137.5	10	1E+30
\$E\$20	Number tested on line 2 Model 4	475	0	147.5	1.136363636	2.083333333
\$F\$20	Number tested on line 2 Model 5	0	-23.75	138.5	23.75	1E+30
\$G\$20	Number tested on line 2 Model 6	1000	0	163.5	1E+30	1.25
\$H\$20	Number tested on line 2 Model 7	0	-6.375	160	6.375	1E+30
\$I\$20	Number tested on line 2 Model 8	0	-2.5	174.5	2.5	1E+30

$$x_{g2}^* = 0$$

- If the unit margin is greater than 174.5 – (–2.5) = 177 USD, then $x_{g2}^* > 0$

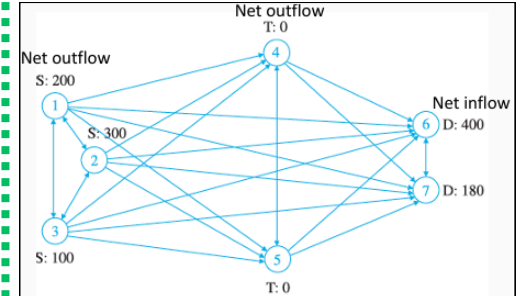
	Wheat	Corn	Sugar Beets
Yield (Tons/acre)	2.5	3	30
Planting cost (\$/acre)	170	150	36
Harvesting cost (\$/T)	200	210	10
Planting price (\$/T)	200	210	10
Harvesting price (\$/T)	200	210	10
Maximum acreage (acres)	200	200	200
Minimum revenue (\$)	0	0	0
Minimum cost (\$)	0	0	0

1	plant	wheat	corn	sugar_1	sugar_2	
2	sell	120	80	300	0	
3	plant	100	0	6000	0	
4	buy	0	0	0	0	
5						
6	yield	2.5	3	20		
7	plant_cost	150	230	260		
8	sell_price	170	150	36	10	
9	buy_price	238	210			
10						
11	sell	233000		500	<=	500
12	buy	0		-200	<=	-200
13	cost	114400		-240	<=	-240
14	profit	118600		6000	<=	6000
15				6000	<=	6000

Set Objective:		\$B\$14
To:	☒ Max ☐ Min ☐ Value Of:	
By Changing Variable Cells:		
\$B\$2:\$E\$4		
Subject to the Constraints:		
\$D\$11:\$D\$15 <= \$F\$11:\$F\$15		

	A	B	C	D	E	F	G	H
1								
2	percent_port	0.44						
3	cost	5	2.4	3.5	5.9	6.9	4.5	3
4	return	5.6	2.7	3.9	6.8	7.7	5.1	3.3
5	percent_return	12.0%	12.5%	11.4%	15.3%	11.6%	13.5%	10.0%
6								
7								
8	profit	\$31.2						

	A	B	C	D	E	F	G	H
1								
2	invest_1	invest_2	invest_3	invest_4	invest_5	invest_6	invest_7	
3	0.44							
4	1							
5	0							
6	1							
7	0							
8								
9								
10								
11								
12								
13								
14								
15								



- Net inflow** = (total inflow) – (total outflow)
- Net outflow** = – (Net inflow)

Example: For Node 4

Total outflow $x_{45} + x_{46} + x_{47}$

Total inflow $x_{14} + x_{24} + x_{34} + x_{54}$

Net outflow

$$= -(x_{14} + x_{24} + x_{34} + x_{54}) - (x_{45} + x_{46} + x_{47})$$

$$= (x_{45} + x_{46} + x_{47}) - (x_{14} + x_{24} + x_{34} + x_{54})$$

- An origin is a node with positive net outflow
- A destination is a node with positive net inflow, i.e., negative net outflow
- A transshipment point is a node with net outflow (and net inflow) equal to 0.

$$\min \quad 5x_{12} + 3x_{13} + \dots + 7x_{76}$$

$$(x_{12} + x_{13} + x_{14} + x_{15} + x_{16} + x_{17}) - (x_{21} + x_{31}) \leq 200 \quad [\text{Node 1}]$$

$$(x_{21} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27}) - (x_{12} + x_{32}) \leq 300 \quad [\text{Node 2}]$$

$$(x_{31} + x_{32} + x_{34} + x_{35} + x_{36} + x_{37}) - (x_{13} + x_{23}) \leq 100 \quad [\text{Node 3}]$$

$$(x_{14} + x_{24} + x_{34} + x_{54}) - (x_{45} + x_{46} + x_{47}) = 0 \quad [\text{Node 4}]$$

$$(x_{15} + x_{25} + x_{35} + x_{45}) - (x_{54} + x_{56} + x_{57}) = 0 \quad [\text{Node 5}]$$

$$(x_{16} + x_{26} + x_{36} + x_{46} + x_{56} + x_{76}) - x_{67} \geq 400 \quad [\text{Node 6}]$$

$$(x_{17} + x_{27} + x_{37} + x_{47} + x_{57} + x_{67}) - x_{76} \geq 180 \quad [\text{Node 7}]$$

$$0 \leq x_{ij} \leq 200$$

	A	B	C	D	E	F
1	Origin	Destination	Unit Cost	Flow		Arc Capacity
2	1	2	2	5	0	200
3	1	3	3	3	180	200
4	1	4	5	0	0	200
5	1	5	5	0	0	200
6	1	6	6	20	0	200
7	1	7	7	20	0	200
8	2	1	1	9	0	200
9	2	3	3	9	0	200

Set Objective:						\$I\$10
To:	☐ Max ☒ Min ☐ Value Of:					
By Changing Variable Cells:						
\$D\$2:\$D\$27						
Subject to the Constraints:						
\$D\$2:\$D\$27 <= \$F\$2:\$F\$27						
\$K\$2:\$K\$8 <= \$M\$2:\$M\$8						
H	I	J	K	L	M	
node	out	in	total		limit	
1	180	0	180	>=	200	
2	300	0	300	>=	300	
3	280	180	100	>=	100	
4	200	200	0	=	0	
5	200	200	0	=	0	
6	180	580	-400	>=	-400	
7	0	180	-180	>=	-180	
cost	3260					

	A	B	C	D	E	F	G
1	Input data						
2	Initial inventory	5000					
3	Holding cost as % of prod cost	5%					
5	Month	1	2	3	4	5	6
6	Production cost/unit	\$12.50	\$12.55	\$12.70	\$12.80	\$12.85	\$12.95
7	Holding cost/unit	\$0.625	\$0.628	\$0.635	\$0.640	\$0.643	\$0.648
8	Production plan (all quantities are in 100s of footballs)						
9	Month	1	2	3	4	5	6
10	Units produced	5000	20000	30000	30000	25000	10000
11		>=	>=	>=	>=	>=	>=
12	Production capacity	30000	30000	30000	30000	30000	30000
16	Demand	10000	15000	30000	35000	25000	10000
18	Ending inventory (>=0)	0	5000	5000	0	0	0
19		>=	>=	>=	>=	>=	>=
20	Storage capacity	10000	10000	10000	10000	10000	10000
22	Summary of costs (all costs are in hundreds of dollars)						
23	Month	1	2	3	4	5	6
24	Holding cost	0	3137.5	3175	0	0	0
25	Production cost	62500	251000	381000	384000	321250	129500
26							1,535,562.50

- Input parameters**
- n = number of decision periods
 - p_t = unit production cost in period t
 - h_t = unit holding cost in period t
 - d_t = demand in period t
 - κ = warehouse capacity
 - u = production capacity
 - i_0 = initial inventory

- Decision variable**
- x_t = production quantity in period t
 - I_t = inventory at end of period t

$$\min \sum_{t=1}^n (p_t x_t + h_t I_t)$$

$$I_t = I_{t-1} + x_t - d_t \text{ for } t = 2, \dots, n$$

$$I_1 = i_0 + x_1 - d_1$$

$$0 \leq I_t \leq \kappa \text{ for } t = 1, 2, \dots, n$$

$$0 \leq x_t \leq u \text{ for } t = 1, 2, \dots, n$$

	A	B	C	D	E	F
1	x1	x2	x3	x4	x5	
2	fix cost	1500	1200	1600	1500	1600
3	labor hour	2	1	6	4	8
4	cloth use	3	2.5	4	4.5	5.5
5	sell price	35	40	65	70	110
6	cost price	20	10	25	30	35
7	revenue	15	30	40	40	75
9		x1	x2	x3	x4	x5
10	produce?	0	1	0	0	1
11	# produce	0	965.52	0	0	379.31
12		<=	<=	<=	<=	<=
13		0	1000000	0	0	1000000

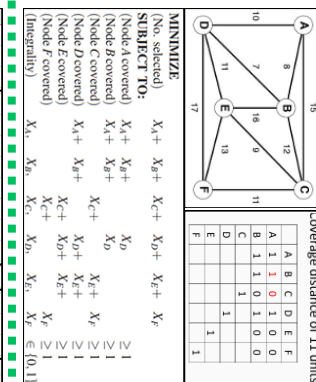
- Decision variable**
- Let $y_j = \begin{cases} 1 & \text{we produce type } j \\ 0 & \text{otherwise} \end{cases}$
 - Let x_j = Number of types- j units to be produced; $j = 1, 2, \dots, 5$
- Objective function**
- $$\text{Max } [(35 - 20)x_1 - 1500y_1] + \dots + [(110 - 35)x_5 - 1600y_5]$$
- Constraints**
- $$2x_1 + \dots + 8x_5 \leq 4000$$
- $$3x_1 + \dots + 5.5x_5 \leq 4500$$
- $$x_j \geq 0; j = 1, 2, \dots, 5$$
- $$y_j \in \{0, 1\}; j = 1, 2, \dots, 5$$

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:



	A	B	C	D	E	F	G	H	I	J
1										
2	decis var	0	0	0	1	0	1	2		
4	A	1	1	0	1	0	0	1	>=	1
5	B	1	1	0	1	0	0	1	>=	1
6	C	0	0	1	0	1	1	1	>=	1
7	D	1	1	0	1	1	0	1	>=	1
8	E	0	0	1	1	1	0	1	>=	1
9	F	0	0	10	0	0	1	1	>=	1
11	fix cost	1	1	1	1	1	1	1		
12	>> (a1 * x1) บังคับให้ 1 site ต้อง cover อย่างน้อย 1 จุด (ถ้าไม่ใส่ x1 จะเลือกเป็น 0 ทั้งหมด)									

Set Objective:

☐ Max ☒ Min

By Changing Variable Cells:

Subject to the Constraints:

Day of Week										
Monday										17
Tuesday										13
Wednesday										15
Thursday										19
Friday										14
Saturday										16
Sunday										11
Input variables										Minimum required number of workers each day
Decision variables (changing cells)										Number of employees working each of the five-day shifts (defined by their first day of work)
Objective cell										Total number of employees on the payroll
Other calculated variables										Number of employees working each day
Constraints										Employees working >= Employees required

	A	B	C	D	E	F	G	H	I	J	K
1	shift start	Mon	Tue	Wed	Thurs	Fri	Sat	Sun			
2	decis var	7	1	3	6	1	5	0	23		
4	Mon	1	1	1	1	1	0	0	18	>=	17
5	Tue	0	1	1	1	1	1	0	16	>=	13
6	Wed	0	0	1	1	1	1	1	15	>=	15
7	Thurs	1	0	0	1	1	1	1	19	>=	19
8	Fri	1	1	0	0	1	1	1	14	>=	14
9	Sat	1	1	1	0	0	1	1	16	>=	16
10	Sun	1	1	1	1	0	0	0	17	>=	11
12	fix cost	1	1	1	1	1	1	1			

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

Problem statement

Revisit the multi-period production problem over an infinite planning horizon.

- Demand rate is constant at 20,000 units/month or $d = (12)(20000) = 240,000$ units/yr. $\rightarrow d = 240000$
- Fixed setup cost $K = 15000$ THB/batch $\rightarrow K = 15000$
- Unit holding cost 0.64 THB/unit/month or $h = (12)(0.64) = 7.68$ THB/unit/year. $\rightarrow h = 7.68$
- What batch size Q minimizes the average annual cost?

We formulate the classical economic order quantity (EOQ) model as a simple nonlinear optimization problem with a single decision variable.

Model formulation

$$\min_{Q>0} f(Q) = K \frac{d}{Q} + h \frac{Q}{2}$$

The stationary point where $f'(Q^*) = 0$ is $Q^* = \sqrt{\frac{2Kd}{h}} = \sqrt{\frac{2(15000)(240000)}{7.68}} = 3061.6$

Note that $f''(Q) = 2Kd/Q^3 > 0$. The objective function is convex (see appendix); therefore, every local minimum is also a global minimum. Because f is strictly convex, Q^* is the unique global minimizer.

Single item

demand rate, d (unit/year)	240000
Setup cost, K (THB/order)	15000
Unit holding cost, h (THB/unit/year)	7.68
Q	60000
Order frequency (order/year), d/Q	4
Order inv. unit, Q/2	30000
Avg setup cost (THB/year), Kd/Q	60000
Avg holding cost (THB/year), hQ/2	230400
Avg total cost (THB/year), Kd/Q + hQ/2	290400

	A	B	C	D	E	F	G
1	customer	lat	long	#shipment	x0-x1	y0-y1	distance
2	1	5	10	200	4.31	-4.97	6.58
3	2	10	5	150	-0.69	0.03	0.69
4	3	0	12	200	9.31	-6.97	11.63
5	4	12	0	300	-2.69	5.03	5.70
6	X	9.31	5.03				
7	total distance	5456.54					

Set Objective:

☐ Max ☒ Min

By Changing Variable Cells:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Method:

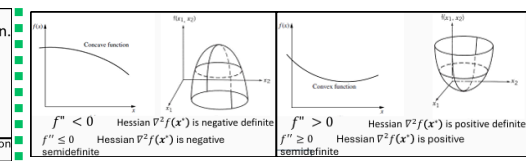
	A	B	C	D	E	F	G	H	I	J
1	customer	lat	long	#shipment	x0-x1	y0-y1		S	T	
2	1	5	10	200	5	-5		5	3	
3	2	10	5	150	0	0		0	0	
4	3	0	12	200	10	-10		7	10	
5	4	12	0	300	-2	2		5	5	
6	X	10	5							
7	total distance	7500								

min $|x|$ $\rightarrow$ $\min_{x,y} \sum_j (|x_j - x| + |y_j - y|)$

min $|f_1(x)| + |f_2(x)|$ $\rightarrow$ $\min_{x,y} \sum_j (|f_1(x_j) - x| + |f_2(y_j) - y|)$

min $|x|$ $\rightarrow$ $\min_{x,y} \sum_j (|x_j - x| + |y_j - y|)$

min $|f_1(x)| + |f_2(x)|$ $\rightarrow$ $\min_{x,y} \sum_j (|f_1(x_j) - x| + |f_2(y_j) - y|)$



Convex quadratic programming (CQP)

$$\max/\min f(x) = \frac{1}{2} x^T Q x + c^T x: Ax \leq b$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric.

- For maximization, Q is negative semidefinite, so $f(x)$ is concave $\rightarrow$ CQP
- For minimization, Q is positive semidefinite, so $f(x)$ is convex $\rightarrow$ CQP
- This can be formulated as SOCP by reformulating the objective function as a constraint.

Example: Hessian of quadratic form

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Consider the following quadratic form

$$q(x) = -2x_1^2 + (-1)x_2^2 + 1x_1x_2 = x^T B x \text{ where } B = \begin{bmatrix} -2 & 1/2 \\ 1/2 & -1 \end{bmatrix}$$

The Hessian matrix $\nabla^2 q(x) = \begin{bmatrix} \frac{\partial^2 q}{\partial x_1^2} & \frac{\partial^2 q}{\partial x_1 \partial x_2} \\ \frac{\partial^2 q}{\partial x_1 \partial x_2} & \frac{\partial^2 q}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix}$

Example: Advertising — Two Solution Methods

Solution 1: Matrix notation

Write $f(x) = x^T B x + a^T x$, where $B = \begin{bmatrix} -2 & 1/2 \\ 1/2 & -1 \end{bmatrix}$, $a = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$.

Using the identity on the previous slide, $\nabla f(x) = (B + B^T)x + a$.

Since $B = B^T$, the first-order condition is $\begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 8 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Thus, $x^* = \begin{bmatrix} 19/7 \\ 20/7 \end{bmatrix} \approx \begin{bmatrix} 2.71 \\ 2.86 \end{bmatrix}$.

Solution 2: Without matrix notation

Recall that $\frac{d}{dx} x^n = nx^{n-1}$. Differentiate with respect to each variable

$\frac{\partial f}{\partial x_1} = -4x_1 + x_2 + 8$

$\frac{\partial f}{\partial x_2} = x_1 - 2x_2 + 3$

Set both partial derivatives equal to zero:

$-4x_1 + x_2 + 8 = 0$

$x_1 - 2x_2 + 3 = 0$

Solving gives $x_1^* = \frac{19}{7} \approx 2.71$, $x_2^* = \frac{20}{7} \approx 2.86$.

It costs a monopolist 5 USD/unit to produce a product. If he produces x units of the product, then each can be sold for $(10 - x)$ USD where $0 \leq x \leq 10$. To maximize profit, how much should the monopolist produce?

Let $f(x)$ be the monopolist's profit if he produces x units

$$f(x) = (10 - x)x - 5x = 5x - x^2$$

Extreme candidates

- Points where $a < x < b$ and $f'(x) = 0$. $f'(x) = 5 - 2x$. Thus, a stationary point is $x = 2.5$, and $f(2.5) = 6.25$
- Points where $f(x)$ does not exist. No candidates
- Endpoints a and b of the interval $[a, b]$. $f'(0) > 0$, so $a = 0$ is local minimum; similarly, $b = 10$ is local minimum. Thus, $x = 2.5$ is the only local maximum.

Note that $f''(x) = -2 < 0$, so $f(x)$ is concave. A local optimum solution is a global optimal solution.